

P8X Instruction Set -- Quick Reference (rev D)

Opcodes, mnemonics and cycle counts generated live from `genucode.py` (the microcode source of truth) -- cannot drift from the hardware.

Operands: #imm immediate | addr 16-bit absolute | (Pn) ptr indirect | (Pn)+ post-increment | (no operand) implied. **By=**bytes, **Cy=c**ycles (incl. fetch). **Flags (F):** C carry (active-high: ADD carry-out / SUB,CMP no-borrow A>=B), Z zero, N negative (bit7), V overflow. '-' = none. Signed branches test N^V / Z. **Memory (rev E):** \$0000-1FFF ROM | \$2000-FEFF RAM | \$FF00-FFFF I/O. P0=PC, P3=stack (empty-descending).

JZ/JNZ/JC are aliases of BZ/BNZ/BCP.

Op	Mnemonic	By	Cy	FI	Description	Op	Mnemonic	By	Cy	FI	Description
System						Stack					
\$00	NOP	1	2	-	No operation.	\$70	PHA	1	2	-	Push A onto P3 stack.
\$01	HLT	1	2	-	Halt clock; resume only by reset.	\$71	PLA	1	3	ZN	Pop A from P3 stack.
\$72	CLC	1	2	C	C←0.	16-bit memory (rev D)					
\$73	SEC	1	2	C	C←1.	\$74	PHW addr	3	9	-	Push 16-bit word at addr (hi then lo: lies little-endian at P3+1).
Interrupts (rev C)						\$75	PLW addr	3	10	-	Pop 16-bit word into addr (lo then hi).
\$02	EI	1	2	-	Enable maskable interrupts (IE=1).	\$76	LP#1 addr	3	9	-	P1 = 16-bit word at addr.
\$03	DI	1	2	-	Disable maskable interrupts (IE=0).	\$77	LP#2 addr	3	9	-	P2 = 16-bit word at addr.
\$04	RTI	1	8	-	Return from interrupt: pop flags then PC; re-enable IE.	\$78	MOVW dst,src	5	13	-	16-bit mem->mem: word at src -> dst.
\$08	IRQ	1	10	-	SW interrupt: push PC+flags, vector to \$0808.	\$79	LP#3 addr	3	9	-	P3 = 16-bit word at addr (restore a saved SP).
Load/store						Push the word at P1+d (n then lo). AI: flags from the address add.					
\$10	LDA #imm	2	2	ZN	A←immediate.	\$80	PHW (P1+d)	2	10	CZN	Push the word at P1+d. AI
\$11	LDB #imm	2	2	ZN	B←immediate.	\$8E	PHW (P2+d)	2	10	CZN	Push the word at P2+d. AI
\$12	LDA addr	3	6	ZN	A←byte at addr (absolute).	\$8F	PHW (P3+d)	2	10	CZN	Push the word at P3+d. AI
\$13	LDB addr	3	6	ZN	B←byte at addr (absolute).	Tier A: C-compiler ISA (2026-09, pure microcode; AI = clobbers A)					
\$14	STA addr	3	6	-	byte at addr←A (absolute).	\$38	LDP1 #imm16	3	5	-	P1←imm16 (3 bytes: was the LPL1/LPH1 pair).
\$15	LDA (P1)+	1	2	ZN	A←[P1], P1++.	\$39	LP#2 #imm16	3	5	-	P2←imm16.
\$16	LDA (P2)+	1	2	ZN	A←[P2], P2++.	\$3A	LP#3 #imm16	3	5	-	P3←imm16.
\$17	LDA (P3)+	1	2	ZN	A←[P3], P3++.	\$3C	AODP#3 #imm	2	6	CZN	P3→P3+imm8 (free a frame). AI: flags from the low byte.
\$19	STA (P1)+	1	2	-	[P1]←A, P1++.	\$3D	SUBP#3 #imm	2	6	CZN	P3→P3+imm8 (allocate a frame). AI: flags from the low byte.
\$1A	STA (P2)+	1	2	-	[P2]←A, P2++.	\$90	LDW addr, (P1+P)	14	CZN	word at addr←word at P1+d. AI	
\$1B	STA (P3)+	1	2	-	[P3]←A, P3++.	\$91	LDW addr, (P2+P)	14	CZN	word at addr←word at P2+d. AI	
\$1D	STA (P1)	1	2	-	[P1]←A.	\$92	LDW addr, (P3+P)	14	CZN	word at addr←word at P3+d (local -> memory word). AI	
\$1E	STA (P2)	1	2	-	[P2]←A.	\$94	STW (P1+d), ad#P	14	CZN	word at P1+d←word at addr. AI	
\$1F	STA (P3)	1	2	-	[P3]←A.	\$95	STW (P2+d), ad#P	14	CZN	word at P2+d←word at addr. AI	
\$51	LDA (P1)	1	2	ZN	A←[P1] (P1 kept).	\$96	STW (P3+d), ad#P	14	CZN	word at P3+d←word at addr (memory word -> local). AI	
\$52	LDA (P2)	1	2	ZN	A←[P2] (P2 kept).	\$98	LDW addr, #imm8	8	-	word at addr←imm8 zero-extended (4 bytes).	
\$53	LDA (P3)	1	2	ZN	A←[P3] (P3 kept).	\$99	LDW addr, #imm16	9	-	word at addr←imm16 (5 bytes).	
\$88	LDA (P1+d)	2	7	CZN	A←byte at P1+d (d unsigned 0..255). C from the address add. Z/N from A.	\$9A	AODW dst,src	5	15	CZNV	word a←a+b, 16-bit; C←1 no borrow (a>=b unsigned). AI; Z high byte only.
\$89	LDA (P2+d)	2	7	CZN	A←byte at P2+d.	\$9C	CPMW dst,src	5	15	CZNV	flags from a←b (16-bit), memory unchanged; C←unsigned a>b, BLT/BGE = signed. AI
\$8A	LDA (P3+d)	2	7	CZN	A←byte at P3+d (a stack local).	\$9E	INCW addr	3	9	CZN	word at addr += 1. AI; flags from the low byte.
\$8C	STA (P1+d)	2	9	CZN	byte at P1+d←A. A kept; flags clobbered by the address add.	\$9F	DECM addr	3	9	CZN	word at addr -= 1. AI; flags from the low byte.
\$8D	STA (P2+d)	2	9	CZN	byte at P2+d←A.	\$A0	AODW addr, #imm8	15	CZNV	word a←a+imm8 (zero-ext), 16-bit; C/carry out; Z of the FULL word. AI	
\$8E	STA (P3+d)	2	9	CZN	byte at P3+d←A (a stack local).	\$A1	SUBW addr, #imm8	15	CZNV	word a←a-imm8; C←1 no borrow; Z full word. AI	
ALU (A,B → A)						\$A2	CPMW addr, #imm8	15	CZNV	flags from a←imm8 (16-bit), memory unchanged; Z full word (a←imm). AI	
\$20	ADD	1	2	CZN	A←A+B.	\$A4	LEAW addr, (P1+P)	14	CZN	word at addr←P1+d (the address of a frame local). AI	
\$21	SUB	1	2	CZN	A←A-B.	\$A5	LEAW addr, (P2+P)	14	CZN	word at addr←P2+d. AI	
\$22	AND	1	2	CZN	A←A AND B.	\$A6	LEAW addr, (P3+P)	14	CZN	word at addr←P3+d (address of a stack local). AI	
\$23	OR	1	2	CZN	A←A OR B.	\$B1	AODW addr, #imm6	15	CZNV	word a←a+imm16; C←carry out; Z full word. AI	
\$24	XOR	1	2	CZN	A←A XOR B.	\$B2	SUBW addr, #imm6	15	CZNV	word a←a-imm16; C←1 no borrow; Z full word. AI	
\$25	CMP	1	2	CZN	Flags from A-B; A,B unchanged.	\$B3	CPMW addr, #imm6	15	CZNV	word a←a AND b; Z high byte only. AI	
\$26	INC	1	2	CZN	A←A+1 (B unused).	\$B4	ANDW dst,src	5	15	ZN	word a←a OR b; Z high byte only. AI
\$27	DEC	1	2	CZN	A←A-1 (B unused).	\$B5	ORW dst,src	5	15	ZN	word a←a XOR b; Z high byte only. AI
\$28	SHL	1	2	CZN	A←A<<1, 0→bit0, out→C.	\$B6	XORW dst,src	5	15	ZN	word a←a AND imm8 (high byte cleared); Z full word. ANDW /JZ tests a bit. AI
\$29	SHR	1	2	CZN	A←A>>1, 0→bit7, out→C.	\$B7	ANDW addr, #imm8	15	ZN	word a←a OR imm8 (high byte kept); Z full word. AI	
\$2A	ROL	1	2	CZN	Rotate A left through carry.	\$B8	ORW addr, #imm8	15	ZN	word a←a XOR imm8 (high byte kept); Z full word. AI	
\$2B	ROR	1	2	CZN	Rotate A right through carry.	\$B9	XORW addr, #imm8	15	ZN	word a←a AND imm16; Z full word. AI	
ALU with T (rev C; 2nd operand T, B preserved)						\$BB	ORW addr, #imm16	15	ZN	word a←a XOR imm16 (BFFFF = bitwise NOT); Z full word. AI	
\$80	AODT	1	2	CZN	A←A+T (B preserved).	Control flow					
\$81	SUBT	1	2	CZN	A←A-T (B preserved).	\$40	JMP addr	3	4	-	P0(PC)←addr.
\$82	ANDT	1	2	CZN	A←A AND T (B preserved).	\$41	JSR (P1)	1	9	-	Push return addr. P0←P1.
\$83	ORT	1	2	CZN	A←A OR T (B preserved).	\$42	RTS	1	6	-	Pop return addr from P3 into P0.
\$84	XORT	1	2	CZN	A←A XOR T (B preserved).	\$43	JSR addr	3	10	-	Push return addr. P0←addr.
\$85	CMPT	1	2	CZN	Flags from A-T; A,B unchanged.	\$48	BZ addr	3	4	-	Branch if Z=1. (JZ alias.)
\$86	LDT #imm	2	2	-	T←immediate.	\$49	BNZ addr	3	4	-	Branch if Z=0. (JNZ alias.)
\$87	LDT addr	3	6	-	T←byte at addr (absolute).	\$4A	BCP addr	3	4	-	Branch if C=1 / A>B unsigned. (JC alias.)
						\$4C	JNC addr	3	4	-	Branch if C=0 / A<B unsigned.
						\$A8	JMP rel8	2	14	-	P0←P0+rel8 (2-byte jump; A/flags kept; assembler relax / JMP.R).
						\$A9	BZ rel8	2	14	-	Branch rel8 if Z=1. (JZ.R alias.)
						\$AA	BNZ rel8	2	14	-	Branch rel8 if Z=0. (JNZ.R alias.)
						\$AB	BCP rel8	2	14	-	Branch rel8 if C=1. (JC.R alias.)
						\$AC	JNC rel8	2	14	-	Branch rel8 if C=0.
						Signed branches (rev C; after CMP)					
						\$44	BLT addr	3	4	-	Branch if signed A<B (N^V<1). After CMP.
						\$45	BGE addr	3	4	-	Branch if signed A>=B (N^V<0). After CMP.
						\$46	BLE addr	3	4	-	Branch if signed A<=B (N^V)/Z). After CMP.
						\$47	BGT addr	3	4	-	Branch if signed A>B. After CMP.
						\$AD	BLT rel8	2	14	-	Branch rel8 if signed A<B (N^V).
						\$AE	BGE rel8	2	14	-	Branch rel8 if signed A>=B.
						\$AF	BLE rel8	2	14	-	Branch rel8 if signed A<=B.
						\$B0	BGT rel8	2	14	-	Branch rel8 if signed A>B.
						Pointer registers					
						\$31	LPL1 #imm	2	3	-	P1 low byte←imm.
						\$32	LPL2 #imm	2	3	-	P2 low byte←imm.
						\$33	LPL3 #imm	2	3	-	P3 low byte←imm.
						\$35	LPH1 #imm	2	3	-	P1 high byte←imm.
						\$36	LPH2 #imm	2	3	-	P2 high byte←imm.
						\$37	LPH3 #imm	2	3	-	P3 high byte←imm.
						\$64	INP1	1	2	-	P1→P1+1.
						\$65	INP2	1	2	-	P2→P2+1.
						\$66	INP3	1	2	-	P3→P3+1.
						\$68	DEP1	1	2	-	P1→P1-1.
						\$69	DEP2	1	2	-	P2→P2-1.
						\$6A	DEP3	1	2	-	P3→P3-1.
						\$6E	TAP1L	1	2	-	P1 low←A.
						\$6F	TAP1H	1	2	-	P1 high←A.
						\$60	TAP2L	1	2	-	P2 low←A.
						\$61	TAP2H	1	2	-	P2 high←A.
						\$62	TAP3L	1	2	-	P3 low←A.
						\$63	TAP3H	1	2	-	P3 high←A.
						\$68	TPA1L	1	2	ZN	A←P1 low.
						\$69	TPA1H	1	2	ZN	A←P1 high.
						\$6A	TPA2L	1	2	ZN	A←P2 low.
						\$6B	TPA2H	1	2	ZN	A←P2 high.
						\$6C	TPA3L	1	2	ZN	A←P3 low.
						\$6D	TPA3H	1	2	ZN	A←P3 high.